Nile Nexus at ArGuard Shared Tasks: Leakage-Controlled MARBERTv2–SVM Ensemble for Arabic Harmful-Prompt Classification

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Abstract

Arabic harmful-prompt detection has remained challenging due to dialectal variation, implicit expressions and context-dependent intent. In the ArGuard 2026 Shared Task, we have participated in Subtask B1, a binary safe/unsafe classification task scored by macro-F1. We have identified evaluation leakage in an initial pipeline: the development set has been reused for early stopping, threshold tuning and ensemble optimization, which has inflated development scores while masking poor generalization on the blind test set. To correct this, we have introduced a leakage-controlled protocol with separate fitting, checkpoint-selection and tuning splits and have compared a TF-IDF SVM baseline, MARBERTv2 and a stacking ensemble, selecting the best-performing method based on tuning-set results. We have further applied five rounds of error-driven data augmentation to address recurring failure patterns. Our final system has achieved 0.861 accuracy, 0.878 precision, 0.862 recall and 0.860 macro-F1 on the blind test leaderboard, ranking 5th. Manual error analysis has shown that remaining errors have mainly stemmed from false positives on sensitive but non-harmful content, confirming that leakage-controlled evaluation has been necessary for a reliable blind-test estimate.

1 Introduction

Detecting harmful intent in Arabic prompts submitted to large language models has remained an open problem, since Arabic has encompassed Modern Standard Arabic and numerous dialects and harmful intent has often been disguised through code-switching, indirect phrasing and cultural references.

ArGuard 2026 at ArabicNLP 2026 has run two tracks: hateful meme detection (Track A) and harmful prompt detection (Track B) (Alam et al., 2026). We have entered Track B1, a binary safe/unsafe classification task scored by macro-F1.

We have fine-tuned an Arabic transformer (MARBERTv2) and have compared it with a TF-IDF character n-gram SVM and a stacking ensemble, choosing whichever has scored best on a held-out tuning slice. Our main contribution has been methodological: an earlier pipeline had reused the dev set for early stopping, threshold tuning and ensemble fitting together and had then reported a dev score from that same data. This has produced a dev macro-F1 near 0.99 that has fallen to around 0.60 on the blind test set, reflecting leakage rather than genuine strength.

We have made the following contributions:

- We have split `train.csv` into disjoint slices for fitting, checkpoint selection and tuning, leaving dev untouched until one final report.
- We have run five rounds of error-driven augmentation, adding safe-topic and hard-negative examples from manual review of mistakes.
- We have run a small blind audit of final test predictions to find remaining failure modes.

Implementation details: https://github.com/Mif-taa/arguard-b1.git

2 Related Work

Harmful-content detection has expanded from English-focused resources toward multilingual and dialect-aware settings (Fortuna and Nunes, 2018). Early Arabic research has focused on offensive language and hate-speech detection, where dialect variation and limited annotated resources have posed significant challenges. The OSACT shared tasks (Mubarak et al., 2020) have provided important benchmarks, while AraBERT (Antoun et al., 2020) and ARBERT/MARBERT (Abdul-Mageed et al., 2021) have advanced Arabic-specific pretrained representations; we have used MARBERTv2 for harmful-content classification.

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Recent work has shifted toward Arabic-specific safety evaluation, with AraSafe (Mubarak et al., 2025) and ArabicDialectSafety (Zaghouani et al., 2026) benchmarking dialect-aware content safety. Stacked generalization (Wolpert, 1992) has also been explored for classification robustness, including in Arabic offensive language systems (Saeed et al., 2020); we have accordingly compared TF-IDF SVM, MARBERTv2 and a stacking ensemble. Recent ArGuard research has extended Arabic harmful-content detection to multimodal and prompt-based settings, with AHA-Memes (Kmainasi et al., 2026) and the ArGuard shared task (Alam et al., 2026) addressing Arabic memes and LLM prompts. However, dialect-aware harmful prompt classification remains relatively underexplored, motivating our approach for Track B1.

3 Data Description

The ArGuard 2026 Task B (Subtask B1) release has provided `train.csv`, `dev_with_label.csv`, `dev_without_label.csv` and a blind test file, each row carrying a prompt, id, dialect tag, binary label (safe/unsafe) and, for unsafe rows, a `granular_label` harm category (e.g. self-harm, harassment, hate speech, fraud, bullying). Table 1 has shown the distribution. Both train and dev have

Split	Rows	Safe	Unsafe
train.csv	16,831	3,395	13,436
dev_with_label.csv	2,405	475	1,930
blind test	7,809	3,958	3,851

Table 1: Dataset distribution for Subtask B1.

exhibited a consistent ~80/20 unsafe/safe imbalance, which we have addressed with class-balanced training objectives rather than resampling.

4 Methodology

4.1 Problem Formulation

Subtask B1: The task is to classify an Arabic user prompt into one of two categories. Given a text t, the objective is to learn a mapping function $f(t) \rightarrow \{0, 1\}$, where 0 indicates "Safe" and 1 indicates "Unsafe".

4.2 Preprocessing

We have dropped rows with missing or blank prompts, missing labels, or out-of-vocabulary label values and have removed duplicate ids. The raw train.csv release has contained 16,835 rows; after dropping 4 rows with missing/blank prompts, missing/invalid labels, or duplicate ids, 16,831 rows have remained (Table 1). dev_with_label.csv and the blind test set have required no drops (0 rows removed from each). We have applied a light Arabic normalization identically across all splits: diacritic and tatweel removal, alef-form unification, alef maqsura and hamza-carrier normalization, tā' marbūṭa normalization and whitespace collapsing.

4.3 Leakage-Controlled Splitting

We have stratified `train.csv` into three disjoint slices: `train_fit` (11,781 rows) has been used to fit the SVM and transformer; `train_earlystop` (2,525 rows) has served only as the Trainer's `eval_dataset` for checkpoint selection; `train_tune` (2,525 rows) has served only for threshold search and ensemble fitting. We have reserved `dev_with_label.csv` (2,405 rows) for a single final report, shown in Figure 1.

4.4 Data Augmentation

We have run five rounds of targeted synthetic augmentation on `train_fit` only, motivated by manual review of earlier-iteration misclassifications on `train_earlystop` and `train_tune`, prior to finalizing the templates used here: (1) generic safe topics; (2) sensitive-sounding but safe religion, gender and business topics; (3) creative writing, riddles and reading comprehension; (4) geopolitics, religious conversion and pronoun-reference exercises; and (5) indirectly phrased, non-graphic hard-negative self-harm and harassment prompts. Rounds 1–4 have added safe-class examples targeting false positives from the prior round; round 5 has added unsafe-class examples targeting a missed false negative. None of the augmentation has drawn on `dev_with_label.csv` or the blind test set. Templates and per-round counts are given in Appendix C (Table 5) and Appendix B (Table 4). For the final submission, we have also added `dev_with_label.csv` to `train_fit`, following organizer guidance. The resulting evaluation trade-off is discussed in Section 5.

4.5 Models and Ensemble

We have fine-tuned UBC-NLP/MARBERTv2 for binary sequence classification using Hugging Face Trainer with macro-F1-based best model selection (Table 2). A TF-IDF character n-gram Linear SVM has been trained as a lexical baseline. We have selected MARBERTv2 for its widespread use in dialect-aware Arabic classification tasks; broader

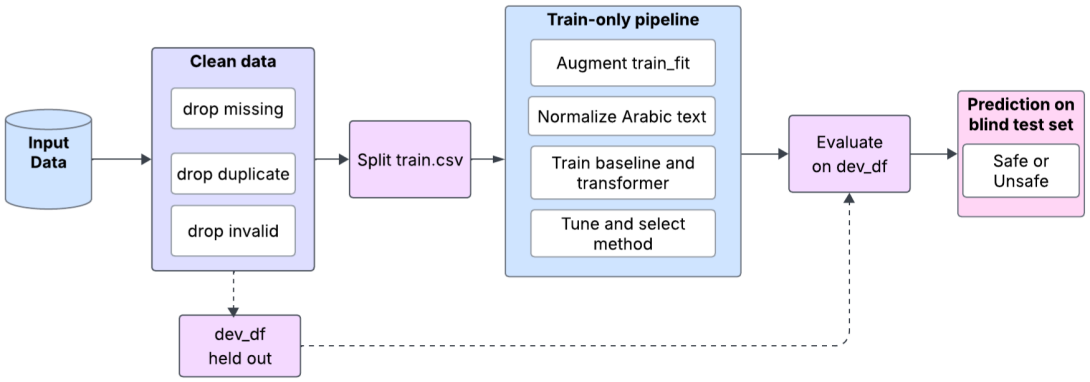


Figure 1: Leakage-controlled data flow: `train.csv` is split three ways; dev is used exactly once.

Hyperparameter	Value
MARBERTv2 fine-tuning	
Max sequence length	128
Batch size	16
Epochs	4
Learning rate	2e-5
Optimizer	AdamW
Weight decay	0.01
Warmup ratio	0.1
Early stopping patience	2 epochs
Seed	42
TF-IDF + Linear SVM	
N-gram range (char)	2–5
Max features	50,000
Class weighting	balanced
Threshold search	
Search range	[0.30, 0.70]
Minimum improvement	0.005 macro-F1
Stacking ensemble	
Meta-classifier	Logistic Regression
Class weighting	balanced

Table 2: Hyperparameters for the transformer, SVM baseline, threshold search and ensemble.

encoder comparison, including AraBERT, has been left for future work. Using `train_tune`, we have optimized the transformer decision threshold and have trained a logistic-regression stacking ensemble over [SVM score, P(unsafe)]. The final blind-test model has been selected solely by `train_tune` macro-F1.

5 Results and Analysis

5.1 Model Comparison on train_tune

Table 3 has compared all methods on `train_tune`: the majority baseline has reached 0.444 macro-F1 (reflecting the 80/20 imbalance), while TF-IDF SVM has reached 0.927 and MARBERTv2 has reached 0.947 at the default threshold. Threshold search has found no candidate beating this by the required 0.005 margin, so the tuned threshold has matched the default. The SVM+transformer ensemble has reached the highest score, 0.949 (a 0.002-point margin over MARBERTv2 alone, plausibly within the noise of the 2,525-row `train_tune` slice) and has been selected as the best-performing method, also separating classes cleanly (Figure 2, ROC-AUC 0.989, average precision 0.997).

Method (train_tune)	F1	P	R	Acc
Majority baseline	0.444	0.399	0.500	0.798
TF-IDF + Linear SVM	0.927	0.925	0.929	0.953
MARBERTv2 (default thresh.)	0.947	0.950	0.944	0.966
MARBERTv2 (tuned thresh.)	0.947	0.950	0.944	0.966
SVM + MARBERTv2 ensemble	**0.949**	0.946	0.951	0.967
Blind Test (leaderboard)	0.860	0.878	0.862	0.861

Table 3: Precision, recall, F1 and accuracy on `train_tune` (method selection) and the final blind-test leaderboard result.

5.2 Development Set and Blind Test Performance

In the final submission configuration, dev has been folded into `train_fit`, so the 0.995 macro-F1 on dev has been only a sanity check (Figure 3), not a held-out estimate. The official leaderboard score has been 0.86 macro-F1. The predicted test distribution (60.3% unsafe / 39.7% safe) has differed from training (79.8% / 20.2%); Appendix A revisits this with gold labels.

5.3 Manual Error Analysis

We have blind-labeled 31 sampled test predictions without viewing model outputs. Agreement has reached 74.2%, with 5 false positives and 3 false negatives, showing slight over-flagging of sensitive-but-safe content (Figure 4). False positives have mainly involved folk remedies, religion and medical descriptions, while false negatives have in-

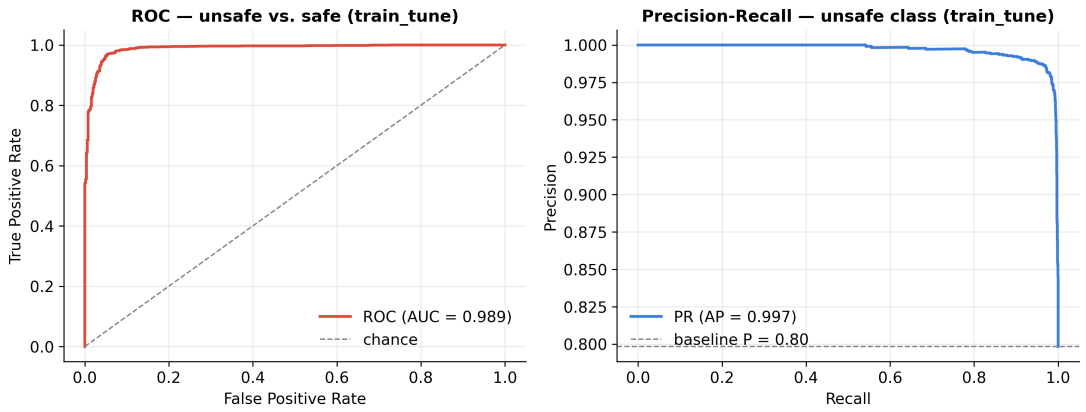


Figure 2: ROC and Precision-Recall curves for MARBERTv2 on `train_tune`.

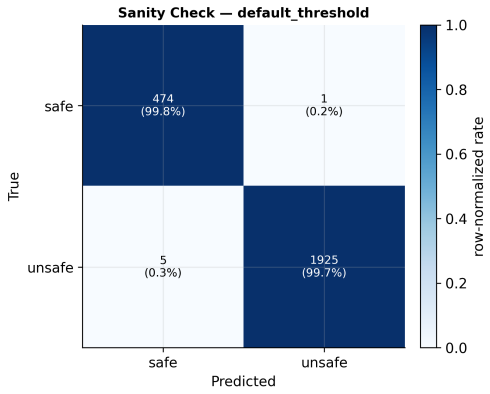


Figure 3: Confusion matrix on `dev_with_label.csv` (sanity check only; see Table 3 for the held-out result).

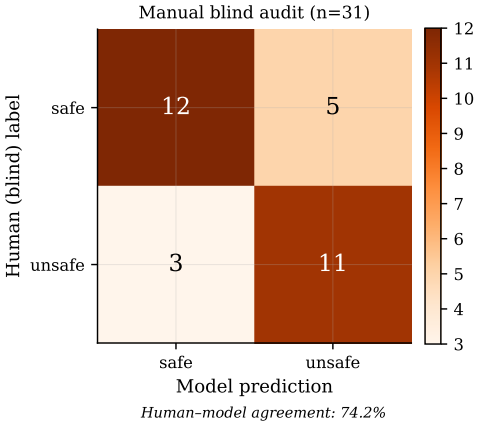


Figure 4: Confusion matrix from the blind manual audit (n=31).

volved indirect self-harm and retaliation framing, indicating remaining gaps in hard-negative coverage.

6 Conclusion

We have presented a MARBERTv2-based system for Subtask B1, combining TF-IDF+SVM and MARBERTv2 via stacking, reaching 0.860 macro-F1 on the blind leaderboard (5th place), selected via 0.949 macro-F1 on `train_tune`. Our main contribution has been methodological: we have restructured the pipeline so checkpoint selection, threshold tuning and ensemble fitting never touch dev, after an earlier version's reuse of dev for all three roles produced an inflated score. We have left k-fold cross-validation and comparison against encoders such as AraBERT for future work.

Limitations

Template-based augmentation has potentially taught patterns rather than concepts. Threshold tuning has used a small `train_tune` subset (2,525 rows) and manual analysis at submission time covered only 31 samples (Appendix A extends this to the full test set). `train_tune` has informed earlier augmentation rounds, so its role in Table 3 is one clean pass on the finalized pipeline, not a wholly untouched split. The binary setup has ignored harm categories and only MARBERTv2 has been tested.

Ethical Statement

Round 5 augmentation has remained non-graphic and has excluded actionable instructions. The blind audit exposed annotators to distressing content. False negatives risk filter bypass; false positives risk restricting legitimate discussion, reflecting reliance on topic cues over intent. The system is treated as a research artifact, not a production tool.

Acknowledgements

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A Full Blind-Test Error Analysis with Gold Labels

The official gold labels for the blind test set (7,809 prompts) have allowed us to replace the 31-sample blind manual audit reported in Section 5.3 with a full, label-verified error analysis, confirming the 0.86 macro-F1 reported in Section 5.

A.1 Confusion Matrix and Error Breakdown

Figure 5 has shown the full breakdown of correct, false-positive and false-negative predictions on the blind test set. The model has therefore erred roughly 8.5 times more often by over-flagging safe content than by missing unsafe content.

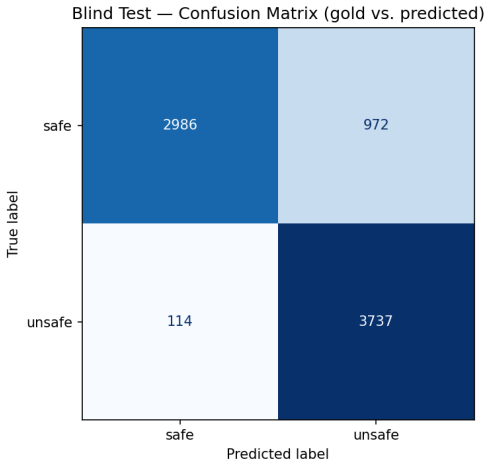


Figure 5: Confusion matrix on the full blind test set (n=7,809).

A.2 Label Distribution and the Dev–Test Gap

The gold labels have shown the blind test set to be close to balanced (50.7% safe / 49.3% unsafe), unlike the ∼80/20 unsafe-majority split of `train.csv` and `dev_with_label.csv`. Since our threshold and ensemble have been tuned on training-distributed data, this class-balance mismatch has been the main driver of the dev–test performance gap.

A.3 Error Rate by Dialect

Figure 6 has broken error rate down by dialect. MSA has made up 68.3% of the test set and has accounted for 937 of the 972 false positives (96.4%).

The false-positive problem has therefore been concentrated in MSA prompts rather than distributed evenly across dialects, consistent with the pattern reported in Section A.4.

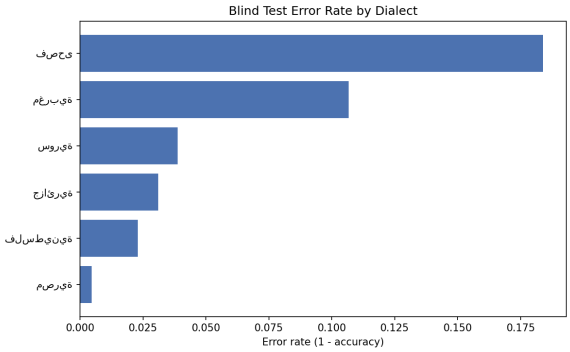


Figure 6: Blind test error rate by dialect.

A.4 Qualitative Error Patterns

False positives have mainly involved neutral factual MSA queries, folk-remedy or household-science text and reading-comprehension prompts featuring a fictional predatory character, confirming the patterns in Section 5.3. False negatives have mainly involved indirectly phrased harassment, victim-blaming framing and conversational self-harm ideation rather than explicit requests for instructions.

B Per-Round Augmentation Counts

Table 4 has reported the number of examples added in each augmentation round described in Section 4.4 and the resulting class balance of `train_fit` after each round.

Round	Added	train_fit size	Safe %
Start (pre-aug.)	–	11,781	20.2
1 (safe topics)	184	11,965	21.4
2 (religion/gender/biz)	76	12,041	21.9
3 (creative/trivia)	34	12,075	22.1
4 (geopolitics/grammar)	62	12,137	22.5
5 (hard-neg. unsafe)	10	12,147	22.5

Table 4: Per-round augmentation counts and resulting `train_fit` class balance.

C Augmentation Templates

Table 5 has listed representative, hand-written (not model-generated) slot-filled templates for each augmentation round (Section 4.4), given in English translation. Round 5 (hard-negative) examples have been described by framing pattern rather than translated, per our Ethical Statement.

Round	Genre	Representative Template (English gloss)
1	Geography / definitions	"What is the capital of Egypt?"
	Rephrase / figurative	"Rephrase the sentence 'Time is like a sword' differently."
	Recipes / products / travel	"How do I cook couscous?"
2	Religion / jurisprudence	"What are the rulings on fasting in Islam?"
	Gender / social topics	"What are the arguments for and against women's rights?"
	Small business	"How do I write attractive marketing copy for a sweets business?"
3	Creative writing	"Write a short story about a trip to the moon."
	Riddles / trivia	"What grows the more you take from it?" (a riddle)
4	Geopolitics	"What are the causes of tension between Iran and Turkey?"
	Religious conversion (comparative)	"What is the religious view on converting to Christianity?" (comparative, non-advocacy)
	Pronoun-reference grammar	"Ahmed got in the car and it was green. What does the pronoun refer to?"
5	Self-harm (indirect framing)	*Not quoted.* First-person expressions of emotional distress and a wish to stop feeling pain, phrased as questions or hedged statements rather than explicit requests – non-graphic, no methods or instructions, matched to the `granular_label` "self-harm."
	Harassment (indirect framing)	*Not quoted.* Scenarios describing repeated unwanted contact or monitoring of a colleague/classmate, framed by the speaker as ambiguous or joking rather than clearly acknowledged as harassment, matched to the `granular_label` "harassment."

Table 5: Representative augmentation templates by round (English gloss). Rounds 1–4 have added hand-written safe-class examples; round 5 has added hard-negative unsafe-class examples, summarized by pattern rather than translated, per our Ethical Statement. All templates have been authored in Arabic; the gloss has not been used in training.